The Security of Biometric Data in Devices with Cancellable Biometrics Technology: A Systematic Review of the Literature

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***Abstract*—In recent years, the use of various biometric technologies has seen exponential growth for biometric recognition and authentication. This expansion is due to the widespread adoption of biometric systems on a day-to-day basis. However, unlike credit cards and passwords, which can be revoked and reissued in the event of a compromise, biometric data is permanently associated with a user and cannot be replaced. This article focuses on the security of biometric data on devices that use cancellable biometrics technology. The crucial importance of protecting biometric information is discussed and a comparison is made with other authentication methods, whether traditional, biological or auxiliary. In addition, it delves into current biometric authentication technologies, highlighting the relevance of various non-invertible or cancellable biometric template protection schemes. These schemes are evaluated based on the risks associated with the security and privacy of biometric data, including identity theft. On the other hand, the use of cancellable methods to evaluate authentication queries and compare biometric information in a protected environment is mentioned. The critical importance of exercising caution when comparing different non-investable biometric template protection schemes is underlined, especially based on reported performance outcomes, given their considerable influence on the security and privacy of biometric data.**

Keywords—Biometric templates, Biometric Cancellable, Cybersecurity, Password, Data Protection, Biometric Sensor.

I. Introduction

The growing use of mobile devices and IoT in everyday life, made people trust more and choose to make transactions and store private and confidential data and information through the internet, which is why the lock screen has become the first security barrier against unauthorized access. since it has standard security mechanisms and in other cases, the use of biometric sensors as a complement to traditional mechanisms to be unlocked. It is for this reason that unauthorized access can result in cases of fraud and data theft, since the mobile device has physical characteristics and traits for the identification of individuals, also known as biometric templates[1].

On the other hand, it should be borne in mind that the protection provided by biometric systems for stored templates is critical, given that biometric data is unique to each person, for example in devices with a fingerprint sensor; By registering the user's fingerprint, you are compromising their unique traits, since it will not change, unlike traditional passwords that can be modified or permanently deleted, that is why, to avoid the theft of biometric patterns and identity, it is crucial that templates become revocable and not invertable, and thus be able to generate cancellable biometric templates. [2][3]

According to cancellable biometrics, it proposes to protect, preserve the privacy and revocability of biometric information through transformations, in such a way that the disclosure of biometric information during authentication is prevented. Some of the most common approaches to mobile biometric authentication are fingerprint scanning and facial recognition, however, there are other physical or behavioral patterns that are unique to each person and can be stored within a biometric template that can be exposed to attack.[4], [5][6]

Therefore, the objective of this systematic review article is to compile information on methods for the creation of cancellable templates to avoid the recovery of the original biometric data.

II. Methodology

The search strategy for the present systematic literature review article was based on a literature analysis including Web-based search engines, using refined English search strings. These are formed using keywords corresponding to the PICO structure.[7]

TABLE I. PEAK TABLE

BEAK	Keyword	Terms
Population (P)	Biometrics	Authentication methods, related to privacy and data protection and voidability.
Intervention (I)	Sensor	Image capture and encoding mechanisms
Comparator (C)	Passwords	Default security mechanisms on devices be PIN, TOKEN, Two-factor authentication
Outcome (O)	Cybersecurity	Related to encryption and stenography of passwords and biometric information on mobile devices

As a result, five research questions were posed for this mapping study

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TABLE II. RESEARCH QUESTIONS TABLE

ID	Research Question
P1	How do you compare to the authentication methods implemented on devices today?
P2	What biometric authentication technologies do you currently have for device and access security?
P3	What are the risks of using biometrics for device access?
P4	What factors ensure confidentiality and privacy applied to biometric authentication?
P5	How could the performance of cancellable biometric templates from cancellable biometrics methods be evaluated?

Search strategy

For the present study, the databases of indexed bibliographies were used: Scopus and IEEE Xplorer. The search for scientific articles in the aforementioned databases was carried out according to the following procedure:

Search strings: Search strings were generated using logical operators: OR for synonyms and AND for combining keywords.

Inclusion criteria: Information related to Cancellable Biometrics, Biometric Authentication, Data Privacy, Data Protection, Biometric Templates, Biometric Identification, Biometric Sensors, Traditional Security Methods, Cryptography and Stenography,

Exclusion criteria: The following articles were omitted.

- Articles published before 2008.
- Items that are blocked from access.
- Non-DOI items.

TABLE III. SEARCH STRINGS TABLE

Database	Search String
SCOPUS	"Cancelable" AND "Biometric templates" AND "Cryptography" AND "Biometric authentication"
	("Biometric authentication" OR "Data privacy" OR "Data Protection" OR "Biometric templates" OR "Biometric identification") AND ("Sensor" OR "Facial detector" OR "Behavior patlern" OR "Speech recognition" OR "cryptography") AND ("Passwords" OR "PIN" OR "Token digital" OR "Two steps verification" OR "Cancellable Biometry") AND ("security level" OR "false positives" OR "Cybersecurity")
IEEE XPLORE	("All Metadata":Cancelable Techniques) AND ("All Metadata":Biometric templates) AND ("All Metadata":Cryptography) AND ("All Metadata":Biometric authentication)
	((("Biometric authentication" OR "Data privacy" OR "Data Protection" OR "Biometric templates" OR "Biometric identification") AND ("Sensor" OR "Facial detector" OR "Petron behavior" OR "Speech recognition" OR "cryptography") AND ("Passwords" OR "PIN" OR "Token digital" OR "Two steps verification" OR "Cancellable Biometry") AND ("security level" OR "false positives" OR "Cybersecurity"))

PRISMA Methodology

Applying the string of words in the databases of indexed bibliographies: Scopus, IEEE Xplore, a total of 82 research papers were obtained, which were submitted to the inclusion and exclusion criteria:[8]

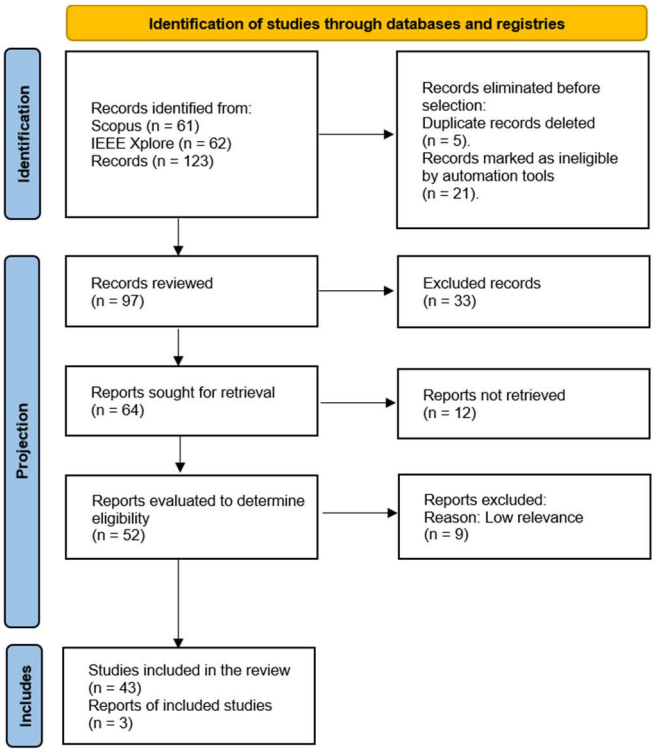

Figure 1. PRISMA Method

III. RESULTS

This section presents the results of each research question and accordingly provides an overview of cancellable biometrics methods or techniques that can be applied in IoT devices.

P1 - How do you compare to the authentication methods implemented on devices today?

The following table will present the research articles found according to systemic mapping, which address issues about the use of security methods that integrate most devices.

TABLE IV. AUTHENTICATION METHODS TABLE

Methods	References	Number of related studies
Traditional Authentication	[1], [9]	3
Biological Authentication	[10], [11], [12]	2
Auxiliary authentication	[10], [13]	2

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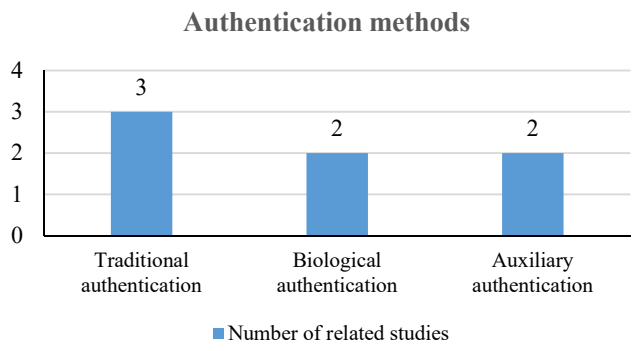

Figure 2. Authentication Methods Found

There are generally three methods of unlocking: traditional authentication, biological authentication, and auxiliary authentication, in which advances have been made related to cryptography and steganography, but there are still shortcomings. [1]

- **Traditional authentication:** These are those known to the device owner, passwords, PIN codes and patterns, which are mostly protected by overly weak user passwords that can be discovered through brute force attacks, or social engineering, which can potentially compromise the integrity of sensitive data.[14]
- **Biological authentication:** Based on the idea of "recognises and unlocks", it can take physiological characteristics as input, be they fingerprints and face, or behavioural characteristics, such as gait and gestures, or simply both, it analyzes them and determines if the user is who they say they are. On the other hand, biometric mechanisms can also use biometric template protection techniques to prevent forgery and increase the security and privacy of information. [15]
- **Auxiliary Authentication:** Based on auxiliary devices is a recent proposal, which uses a smartphone to control the lock/unlock status of other smart devices, i.e. smart doors for data transmission. It is commonly used in smartwatches to verify identity, since they have built-in access to payment method and related device information.[10][13]

Based on the articles found, the following conclusion was reached: in the methods, safety has the same level of identification accuracy obtained according to the method used, however, in few opportunities an analysis is carried out with respect to irreversibility and detachment [6]. Regarding irreversibility, it is the difficulty in reversing the transformations applied to obtain the original biometric template, while the transformations of applied characteristics must be analyzed in detail. On the other hand, in detachment, the different transformed templates are generated from a single biometric template that applies different parameters that have to appear random to themselves.[5]

P2 - What biometric authentication technologies do you currently have for device and access security?

The biometric authentication technologies found according to the applied systemic mapping are classified into the following categories: 1) physiological biometrics (Facial, Fingerprint, Iris), 2) behavioral biometrics (Keystroke dynamics, Voice pattern, body print), and are determined [16]by level of common use.

Below are articles related to biometric technologies for authentication.

Categories	Sensors	References	Number of related studies
Physiological Biometrics	Fingerprint:	[5], [10], [11], [14], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26] [27]	31
	Facial:	[28], [29], [30], [31], [32], [33] [34] [35]	
	Iris:	[4], [5], [36], [37], [38], [39], [40], [41]	
Behavioral Biometrics	Pulsations:	[13],[42], [43]	12
	Voice:	[6] [9], [44] [45] [46]	
	Body print	[12], [47], [48] [49]	

TABLE V. BIOMETRICS CATEGORY TABLE

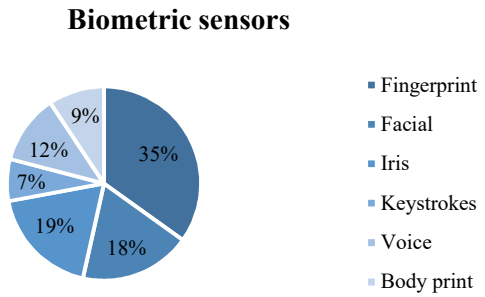

Figure 3. Percentage of sensor usage

As time progresses, biometric technologies seek to ensure the linking of identities to bodies through networks that foster and support identity security in various areas, compared to traditional authentication and identification mechanisms (password, card, pin, and tokens) that are vulnerable to theft and attacks.

In this sense, fingerprint and face biometric technologies proved to be the most suitable for users, since by capturing and encoding images of fingerprints, iris and facial images, vulnerability to attacks caused mostly by weak passwords is reduced. On the other hand, the development of modern biometric systems with cancellable technology, improve the security of the original biometric templates through

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unidirectional transformations and encryption schemes to keep the original biometrics secure and away from use in the system.[31]

Nowadays, the biometric data that is handled in devices for registration or authentication plays an important role, since they are the ones that have the greatest acceptance today and the greatest technological progress is being developed by the brands that manufacture mobile devices, for this reason these authentication mechanisms, whether by fingerprint or face, They could complement traditional methods, such as multi-factor authentication where biometric sensors could be put together and a PIN number entered into a random keypad, in order to prevent shoulder surfing attacks.[48]

P3 - What are the risks of using biometrics for device access?

Biometric data typically incorporates security mechanisms with a degree of encryption for the protection of biometric templates, and mostly based on cryptographic techniques and principles that securely link a cryptographic key to a biometric trait and is stored as a template, in such a way that it does not reveal any information about the biometric key or template. However, these keys can be reconstructed in their entirety only if an accurate query is presented during the identification/verification stage. [6]

Currently, deep learning techniques can artificially synthesize an image, video or audio by realistically exchanging the biometric traits of one user for another, this action is called DeepFakes and threatens applications with biometric systems, since it allows synthesis, the exchange of identities, the manipulation of attributes and the exchange of expressions through learning. However, this variation is usually limited within a small threshold value, and therefore, an imposter will not be able to authenticate correctly since their sample will differ considerably from the stored genuine sample and consequently will not be able to retrieve the correct key.[15]

Another phishing attack occurs when intruders use images, flattened papers and masks, or finger mark patterns for authentication. This could be taken as a false positive, but is currently counteracted by the fact that the advance of cameras and proximity sensors collect more parameters, such as depth, which helps to make a more concise analysis of the face. [50]

P4 - What factors ensure confidentiality and privacy applied to biometric authentication?

Biometric authentication technologies on devices comply with sensitivity factors to increase the level of data security. However, both developers and users must understand that there is no perfect solution for a strong password, since eventually with the advancement of technologies or dedicated algorithms they will be easily decrypted, in turn they will depend on how the data is used and how access control systems must balance the ease of access of legitimate users with the need to defend the system from attacks. during transfer, processing, and storage.[18]

Therefore, in this part, cancellable biometrics mechanisms will be seen as factors that ensure confidentiality, broadly speaking, classification in linking systems and key generation.

TABLE VI. TABLE OF CANCELLABLE METHODS

Schemes	Methods	References
Transformation	Transformations Geometric	[25], [9]
	Random Screenings	[26], [27] , [33], [41]
	BioConvolving	[28], [29]
	BioHashing	[30]–[34] [34], [46] [32]
	Random Permutations	[27], [35], [27], [49]
	Bloom Filters	[36]–[38], [35]

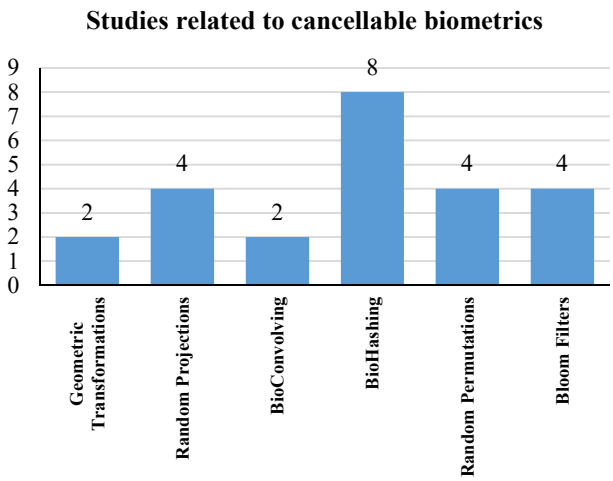

Figure 4. Studies found on cancellable biometrics

Geometric Transformations

Focused on the transformation of the original biometric templates by applying signal domain and feature transformations, which divide and enumerate two-dimensional biometric information into smaller blocks, cells, or regions of data. However, before applying these transformations, the images must be entered by estimating the position and orientation of the singular points (core and delta) and expressing the minutiae in positions and angles in Cartesian or polar coordinates.[19]

The coordinate system is divided into fixed-size cells to randomly change the position of the cells where the renewable information defines the parameters of the transform. It should be noted that this transformation is not a simple permutation since the irreversibility condition requires that cells be mapped to the same cell.

One of the limitations of polar and Cartesian transformations is that they are unstable in the sense that a small change in the position of minutiae in the original digital template can cause a large change in the position of minutiae after transformation if the point crosses a defined boundary.[14]

Random Screenings

The method of transformation by random projections allows the generation of irreversible biometric templates in a

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linear way for the reduction of biometric dimensionality extracted on a random subspace, in which the original information is divided into segments and exchanged randomly, so that it is possible to change the original shape of the data and still retain its statistical characteristics useful for recognition. Since there will still be some outliers that are sensitive to intra-user variations due to low computational complexity[36].

On the other hand, the computational complexity of this protection technique will depend on the probability distribution selected for the projection matrix, for example, when extracting the user's iris pattern, the characteristics are calculated and a different random projection is applied for each application, separated into sectors and resulting transformed vectors are concatenated to form the cancellable template, and then transfer the new pattern into the application's database[40].

Finally, random projections protect authentication and identification systems when the random projection array or transformed patterns are compromised, a new random projection matrix can be created and get a new transformed pattern that can be updated in the application database, on the other hand, biometric templates can be used directly for authentication instead of the original ones without degrading the performance of A classification algorithm based on sparse representations.

BioConvolving

It is a method applicable to any biometric that can represent templates in sequence sets to perform linear convolution of dimensional biometric information. Its application is given by transforming each sequence of the original template into a new sequence using a set of random numbers from 1 to 99, to be subsequently organized into a vector. This process will result in a set of cancellable biometric templates that are unique and cannot be used to reconstruct the original biometric template. The BioConvolving method is a promising technique for the protection of biometric templates, as it is easy to implement and can be applied to a wide variety of biometrics. In addition, the cancellable biometric templates generated by BioConvolving are highly secure and cannot be used to reconstruct the original biometric template, ensuring the privacy and security of the biometric data.[30][45]

BioHashing

BioHashing is a biometric data protection technique that employs random projections and one-way transformations and converts them into an unrecognizable form, ensuring that the generated template cannot be used to reconstruct the original information.[28]

Advanced methods such as the wavelet transform are used to extract the biometric features. This mathematical technique decomposes information into different scales and locations, allowing the identification of unique and relevant patterns in biometric data. Subsequently, the random hash function is applied to these features, as they transform the biometric information into a compact and secure representation, resulting in cancellable, highly secure biometric templates that cannot be used to reconstruct the original biometric feature, ensuring the privacy and security of the biometric data. In addition, BioHashing is an efficient and easy-to-implement technique that can be applied to a wide variety of biometrics.[29][20], [21]

Random Permutations

The random permutation method is a biometric data protection technique that is based on the random permutation of biometric information. This technique is used to protect biometric information and make it cancellable and renewable. It can be applied to one-dimensional or two-dimensional biometric information, dividing it into segments or blocks respectively, and then permuting it randomly. Logical operations such as XOR or XNOR can be used to combine the permuted data, increasing security and privacy, while affecting the recognition rate[40].

The ability to revoke and renew depends on the random number generator used. Matching metrics or techniques in the decision module, such as Euclidean distance and DTW distance, are critical to the protection of these random permutations, as they consider invariant or variable distances to the order of the elements.[37]

These methods gradually reduce the amount of information available for recognition, ensuring that cancellable biometric templates are secure and cannot reconstruct the original template. It is an efficient technique that can be adapted to various biometrics, guaranteeing the security and privacy of biometric data.

Bloom Filter

The Bloom filter method is a probabilistic structure employed to evaluate membership inquiries and compare biometric information in a protected domain, in which the information comparison is directly linked to the ability to revoke and renew, depending on the quality of the random number generator. [39]

For its implementation, the biometric information is divided into blocks of equal size, which establish the bits of the Bloom filter, then assign them to zero and then set them according to the results of an independent binary operation, to which a transformation based on adaptive Bloom filters is applied to vectors of binary characteristics of different recognition algorithms. In this case of Iris, to later generate cancellable biometric templates without the need for alignment.[38]

This transformation offers high security without sacrificing recognition accuracy, enabling the protection and compression of biometric data in a computationally efficient way. However, the effectiveness of this technique is affected by the metrics or matching techniques in the decision module, considering invariant or variable distances to the order of the elements, such as Euclidean distance and DTW distance.[4]

P5 - How could the performance of cancellable biometric templates from cancellable biometrics methods be evaluated?

In this last part, key points to be measured by each cancellable biometrics method will be presented with respect

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to non-investability, cancellability and performance characteristics. Therefore, the methods used to evaluate the extent to which the non-invertible transformations found satisfy these four characteristics will be considered.[23]

Measure of non-investability

It explains that a function can be invertible or non-invertible and describes that an invertible function maps each input to a different output, i.e., the mapping is one-to-one. On the other hand, a non-invertible function maps multiple inputs to the same output, i.e., the mapping is many-to-one. In this case, even if we know the outputs and the function used to generate them, we can't determine the single set of inputs, because the reverse mapping would be one-to-many. Three approaches to measuring non-investibility in transformations are proposed below, taken from the following sources:[22], [24], [25]

Non-invertibility proof: It seeks to demonstrate that the transformation is non-invertible by showing that the direct mapping is many-to-one, implying that the inverse mapping is one-to-many. However, this test does not provide a numerical estimate of non-investability.

Brute Force Estimation: The number of attempts required to decrypt the original template from the brute-force template is estimated. This approach seeks to calculate the complexity of deciphering the original inputs from the protected outputs.

Unrecognizable Percentage: The percentage of the original template that remains unrecognizable if the protected template is compromised is calculated. This approach seeks to determine how much information is lost or remains hidden in the event of unauthorized access to the protected template.

Measure of cancellability

Among the techniques used to evaluate cancellability is the correlation between the templates generated and the ability of an attacker to generate a template from known biometric data, which are usually presented in two approaches to the analysis of cancellability in a template protection scheme, these are obtained from:[23], [26]

Simple Claim of Cancellability Approach: A biometric template protection scheme is claimed to be cancellable due to the possibility of changing the non-invertible transformation or a set of external parameters used in the generation of the protected template.

Empirical Cancellability Assessment Approach: This approach is more prevalent, given that it involves generating multiple protected templates from the same unprotected template and then attempting to match these templates. If the probability of matching is low (similar to the probability of matching protected templates from different unprotected templates), it is concluded that the schema is capable of establishing different identities from the same biometric data, indicating that it is cancellable. This method seeks an empirical test of the schema's ability to generate multiple identities from a single source.

Performance Measurement

The last requirement for a biometric template protection scheme is related to the performance of the recognition system in which they are implemented. It is crucial that the use of protected templates does not significantly degrade the recognition accuracy achievable when making comparisons on the original, unprotected domain. Therefore, the higher the non-invertibility, the less unique the protected template becomes, and therefore the worse the recognition accuracy as mentioned in . Evaluating the performance of a non-invertible biometric template protection scheme follows a standard procedure in the literature, similar to that used to evaluate the performance of a traditional fingerprint recognition system. A fingerprint database is chosen and a protected template is generated for each fingerprint in the database. All protected templates are then compared to each other to calculate the system's False Rejection Rate (FRR) and False Acceptance Rate (FAR).[22], [25][23]

IV. Conclusions

In conclusion, protecting biometric data using traditional techniques such as encryption and secure hashing faces significant challenges due to the variable nature of the data with each new sample. This problem is compounded when encrypted data needs to be decrypted for comparison, or when the hash function does not allow for a direct comparison without compromising security. Faced with these challenges, cancellable biometric techniques emerge as a promising alternative, allowing not only to protect sensitive information but also to guarantee data privacy in authentication systems.

However, evaluating the performance of these cancellable methods remains a critical area that requires meticulous attention. Inconsistencies in the databases and feature extractors used for testing can significantly affect the interpretation of results, suggesting the need for a more uniform standard in performance evaluations. In addition, it is crucial that these assessments consider realistic threat scenarios, including those where an attacker could have access to the transformation and relevant parameters. Only through rigorous evaluation and continuous improvement of these technologies will we be able to move towards more secure and effective biometric authentication systems.

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